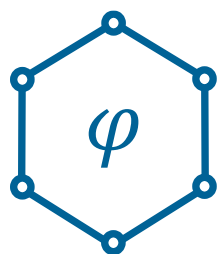


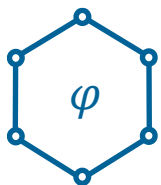
Knowledge Graphs



Emanuel Sallinger



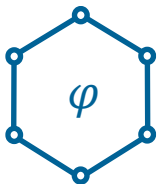
Logical Knowledge in KGs



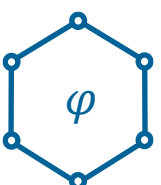
Challenges



Recursion in **SQL**



Recursion in **Datalog**



Recursion in the **Real World**





...

Cypher

SPARQL

Datalog

SQL

First-order Logic

Relational Algebra

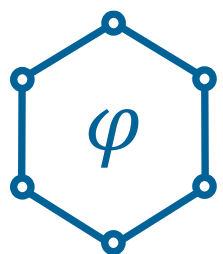
SQL

```
WITH RECURSIVE temp AS (  
    SELECT * FROM flight  
    UNION ALL  
    SELECT t.Origin, f.Destination  
    FROM temp t, flight f  
    WHERE t.Destination = f.Origin  
)  
SELECT * INTO connection FROM temp
```

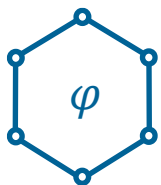
Datalog

```
connection(X,Y) :- flight(X,Y,_).  
connection(X,Z) :- connection(Y,Z), flight(X,Y,_).
```





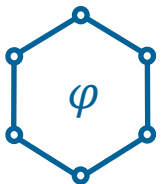
Logical Knowledge in KGs



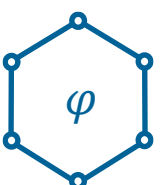
Challenges



Recursion in **SQL**

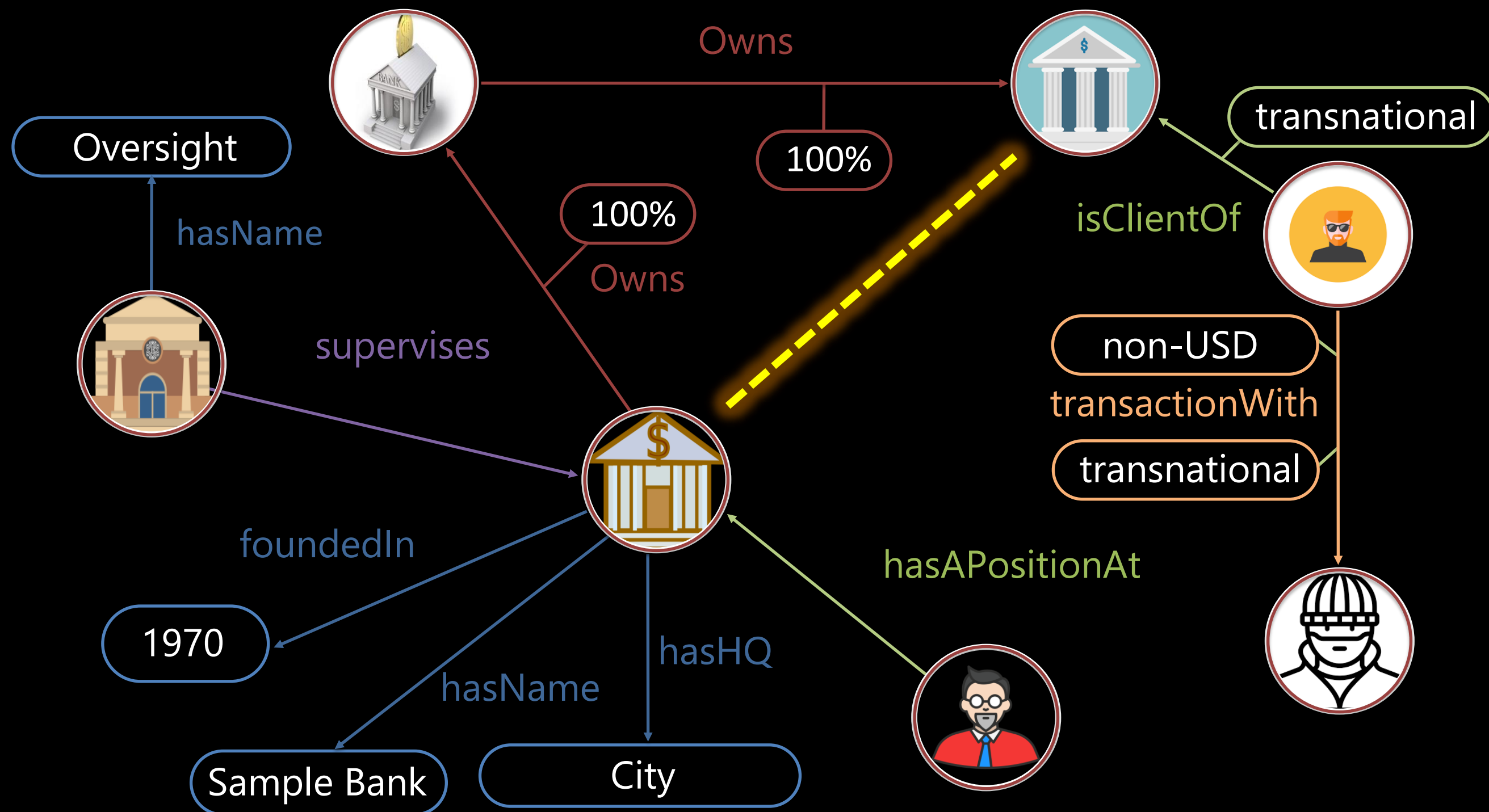


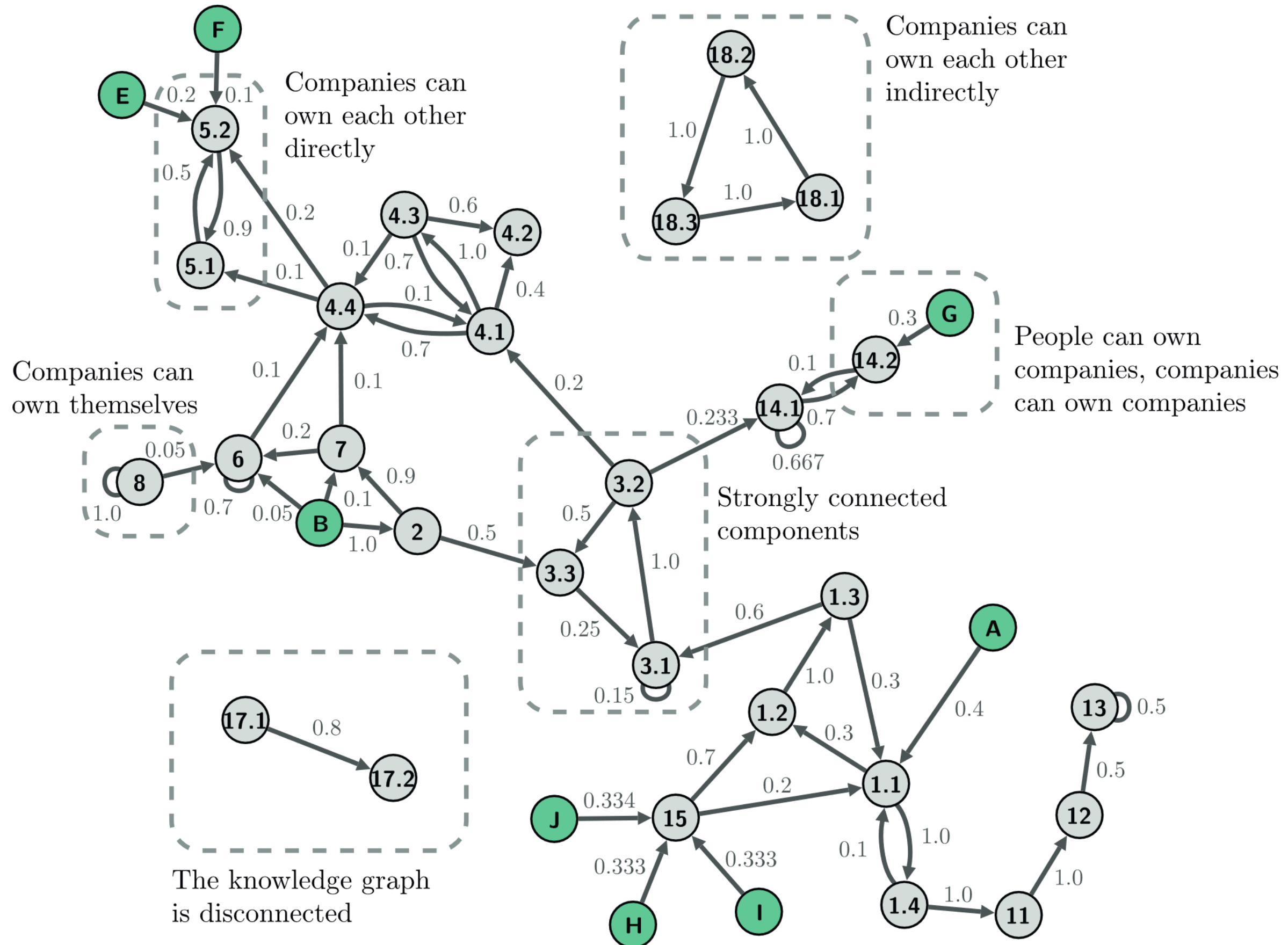
Recursion in **Datalog**

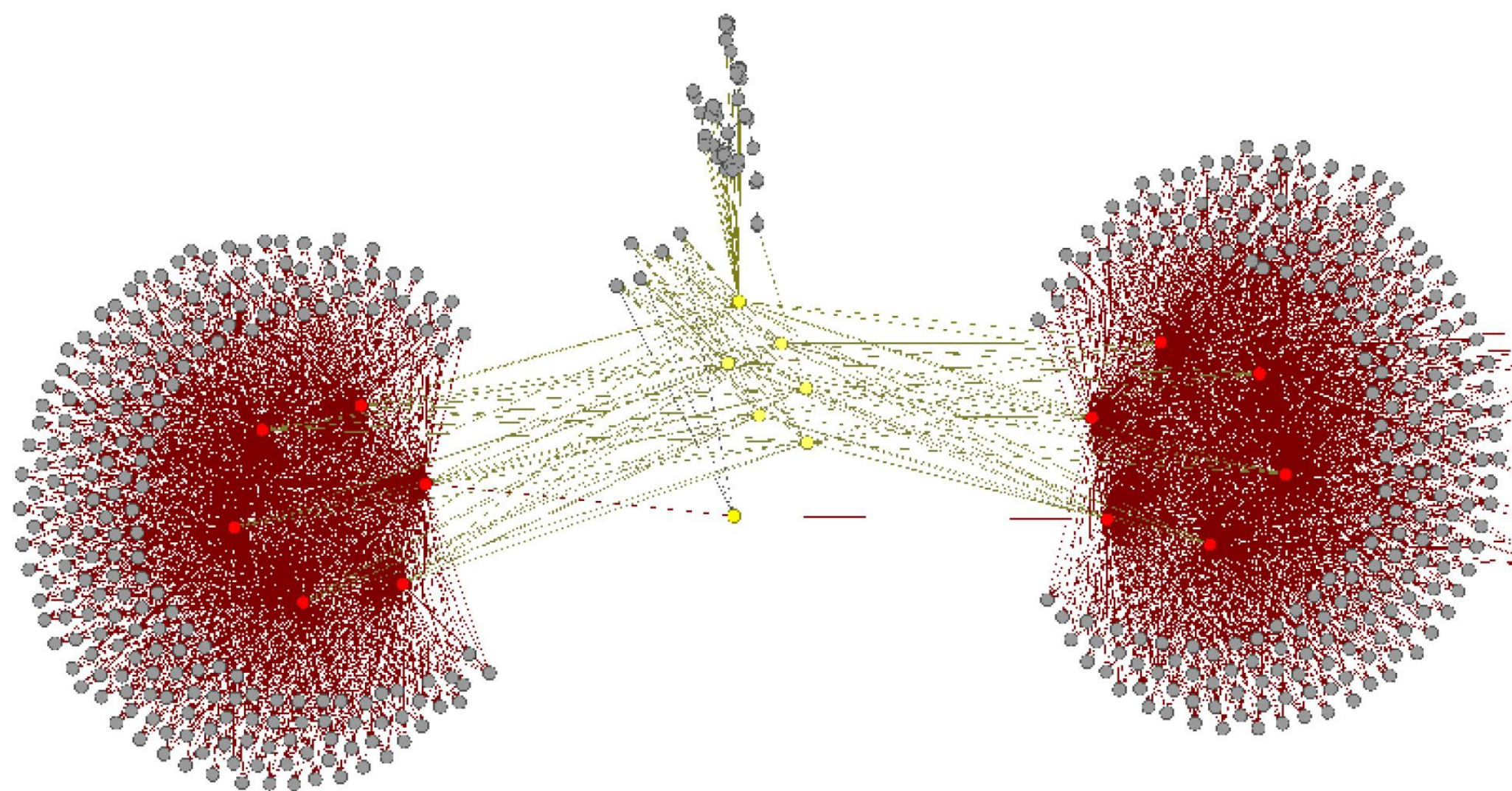


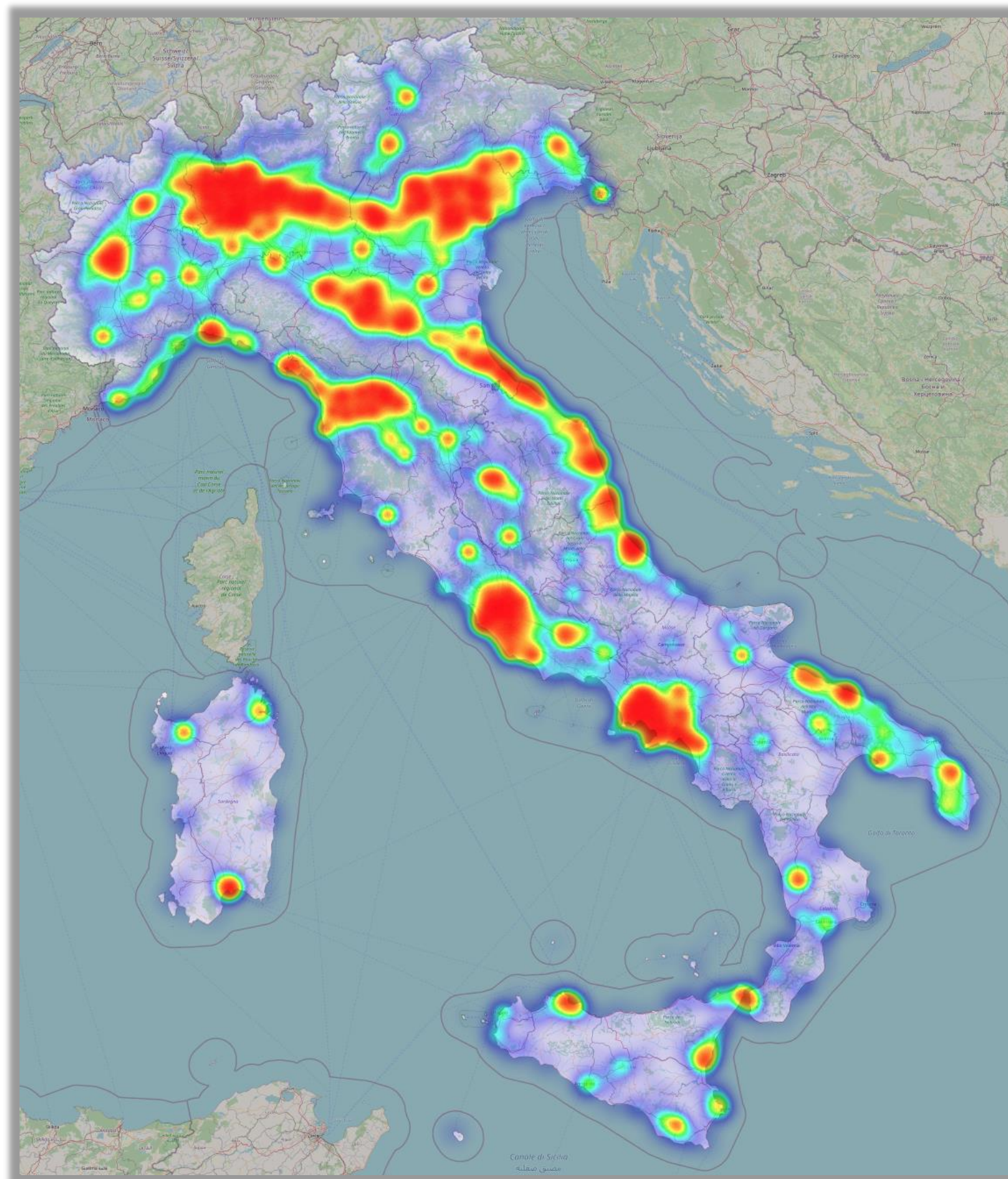
Recursion in the **Real World**



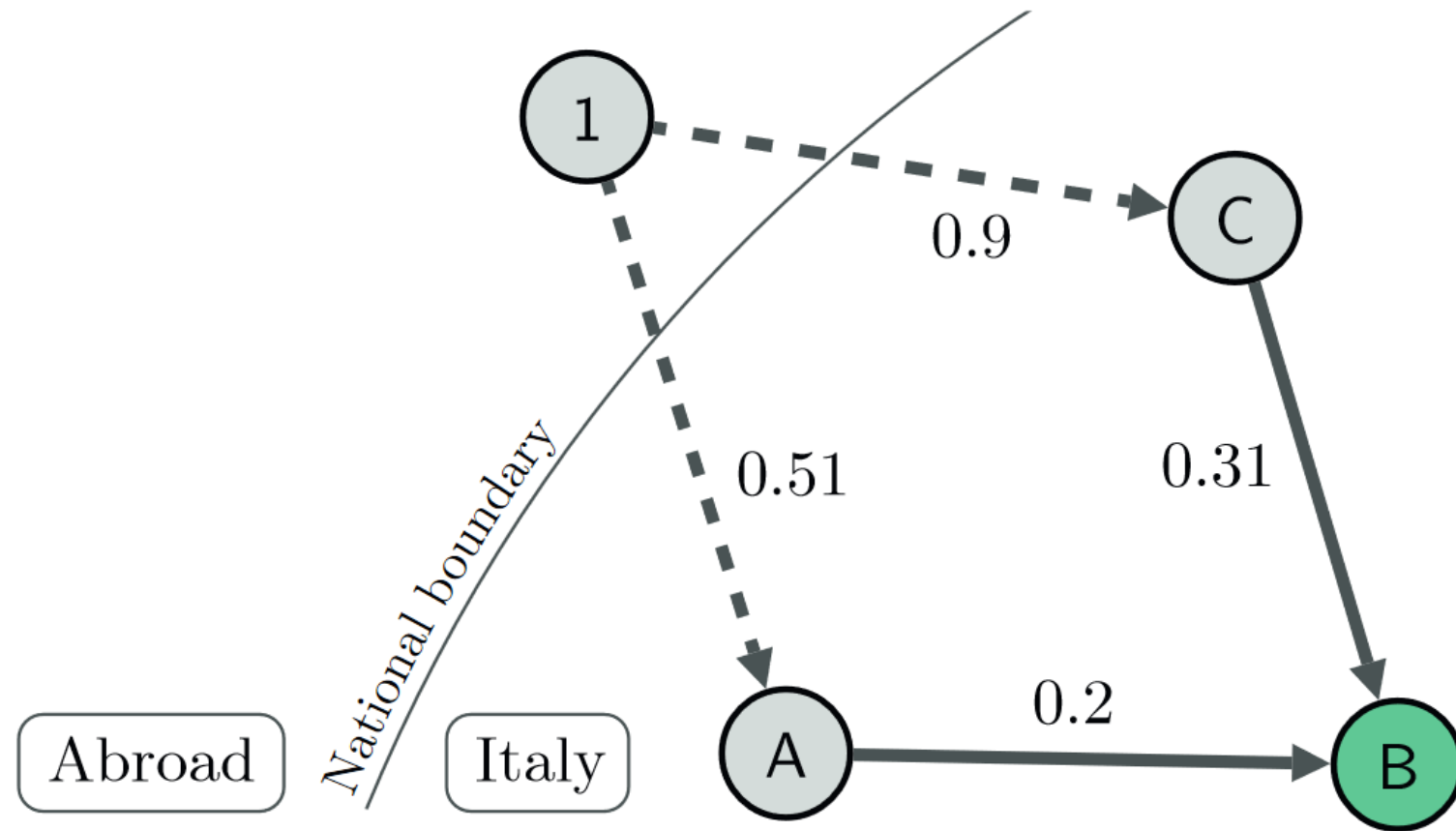








Golden Power Check

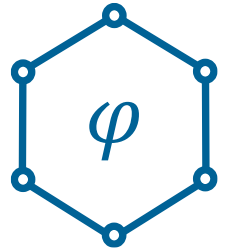


$$T(x, y, w) \rightarrow Own(x, y, w) \quad (1)$$

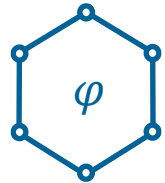
$$Control(x, y) \rightarrow Control(x, x) \quad (2)$$

$$Control(x, y), Own(y, z, w), v = msum(w, \langle y \rangle), v > 0.5 \rightarrow Control(x, z) \quad (3)$$

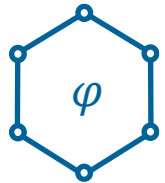
$$F(x), S(y), Control(x, y) \rightarrow GPCheck(x, y) \quad (4)$$



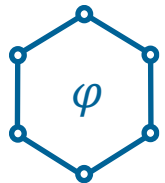
Logical Knowledge in KGs



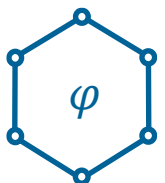
Creation



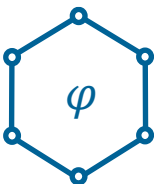
Recursion and Creation
A Unified Framework



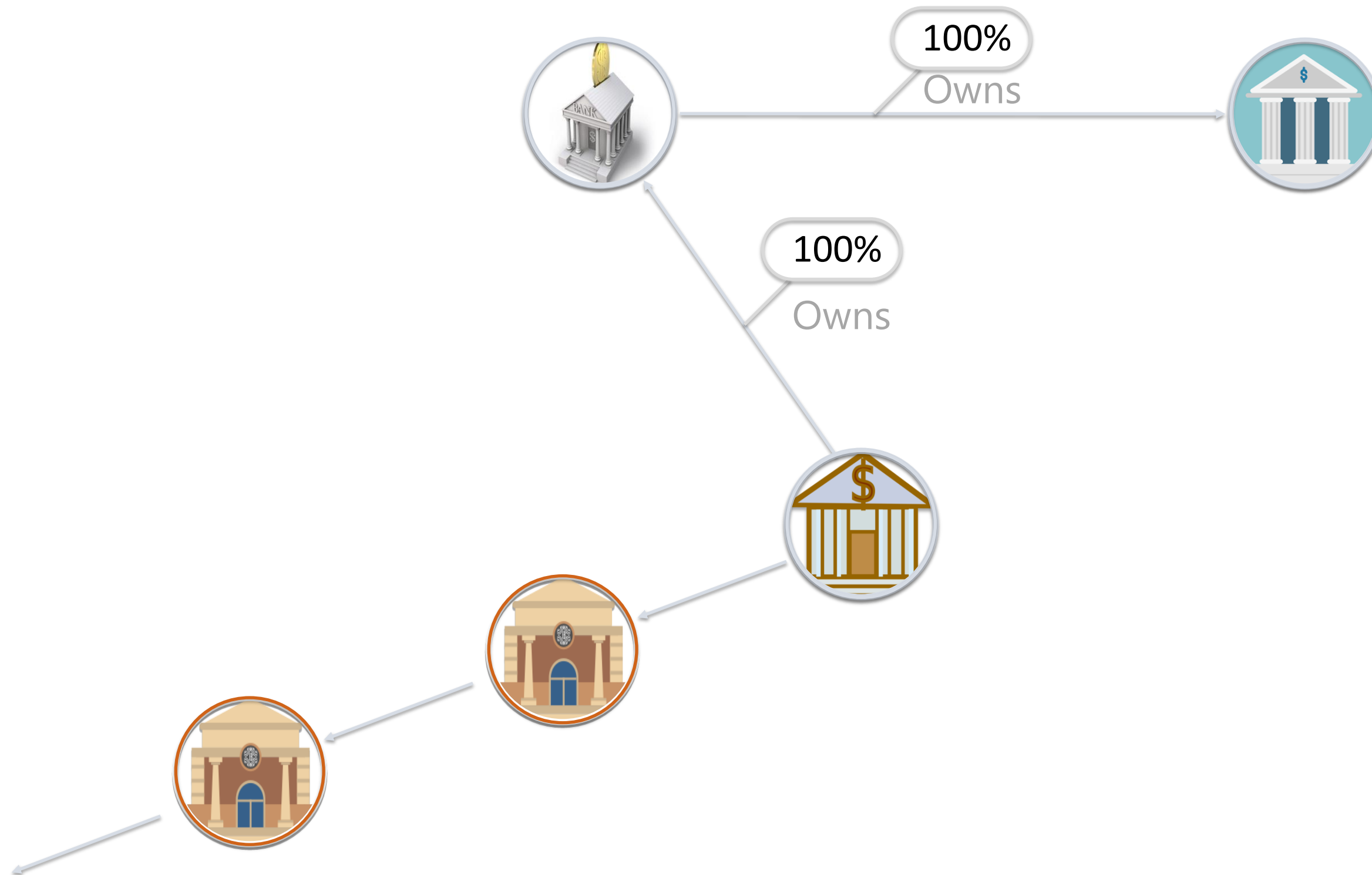
Recursion and Creation
Warded



The **Real World**



Theory to **Practice**





SQL: **NULL**

Recall

v IS [NOT] NULL

- **NULL** indicates an “unknown” value
 - many applications (not applicable, not existent, ...)
 - often misused
- **NULL** is different from any other value, including itself
 - **NULL = NULL** is not true!
 - to test for null, use **IS [NOT] NULL**



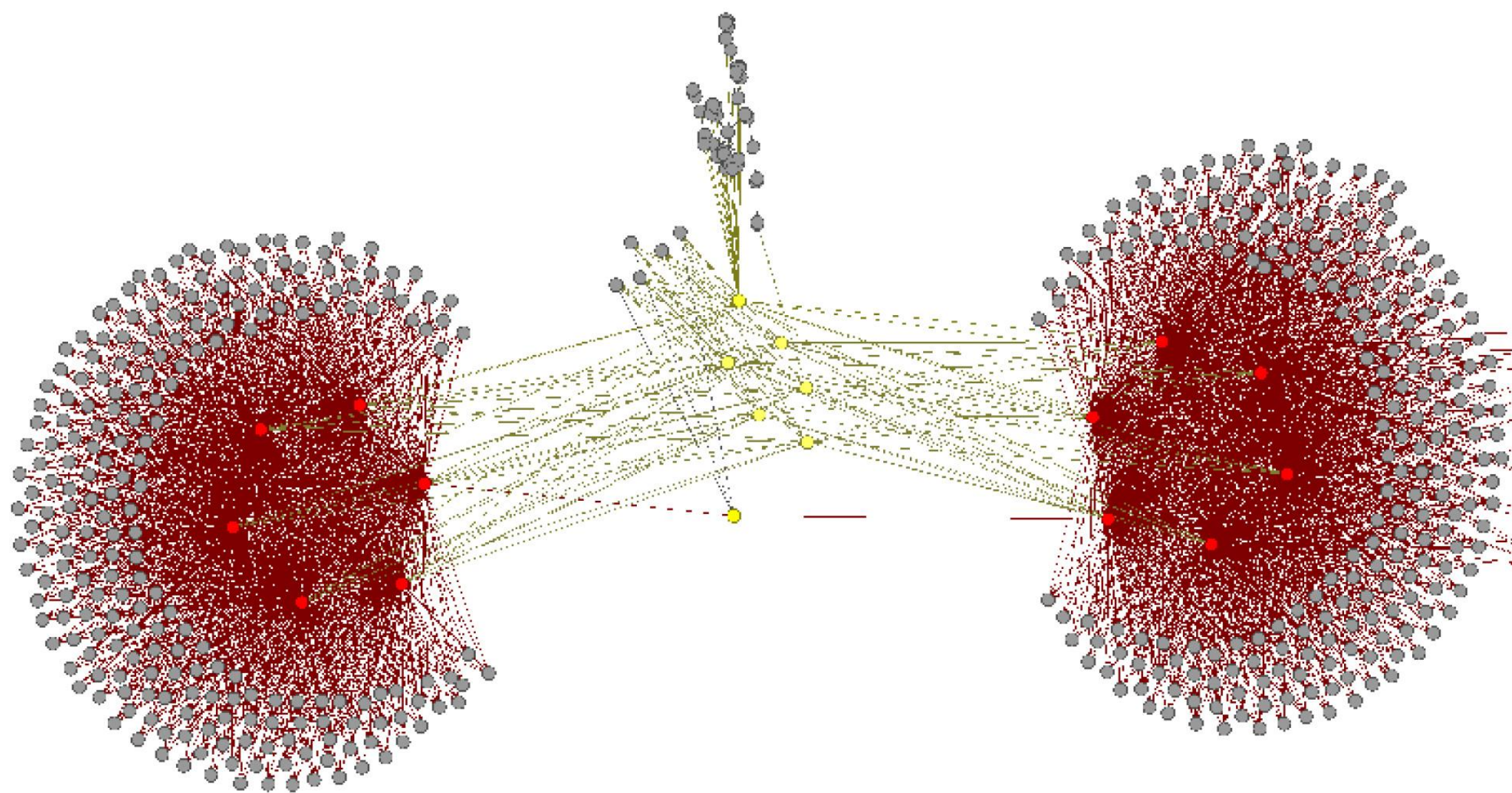
A Unifying Framework

Tuple-Generating Dependencies

$$\forall x \left(\varphi(\vec{x}) \rightarrow \exists \vec{y} \psi(\vec{x}, \vec{y}) \right)$$

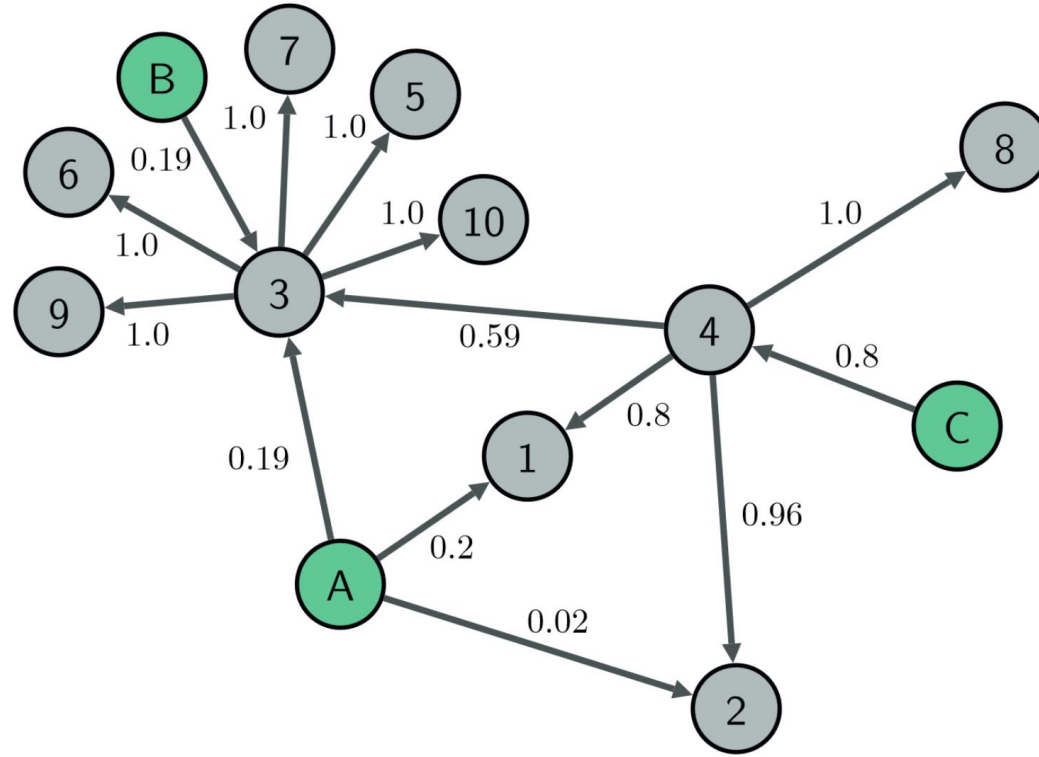
Equality-Generating Dependencies

$$\forall x \left(\varphi(\vec{x}) \rightarrow x_i = x_j \right)$$





Conglomerates



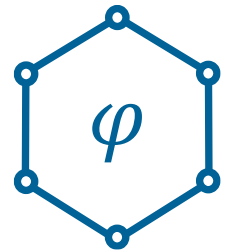
$$O(a, b, x), x > \epsilon \rightarrow U(a, b, x), U(b, a, x) \quad (1)$$

$$U(a, b, x), a > b \rightarrow \exists z \ C(z, a), C(z, b) \quad (2)$$

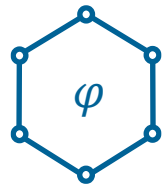
$$U(w, a, x), U(w, b, x) \rightarrow \exists z \ C(z, a), C(z, b) \quad (3)$$

$$Company(a) \rightarrow \exists z \ C(z, a) \quad (4)$$

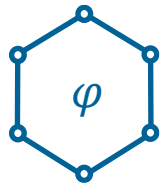
$$C(y, a), C(x, a) \rightarrow x = y \quad (5)$$



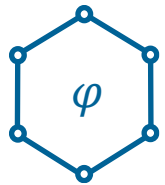
Logical Knowledge in KGs



Creation



Recursion and Creation
A Unified Framework



Recursion and Creation
Warded

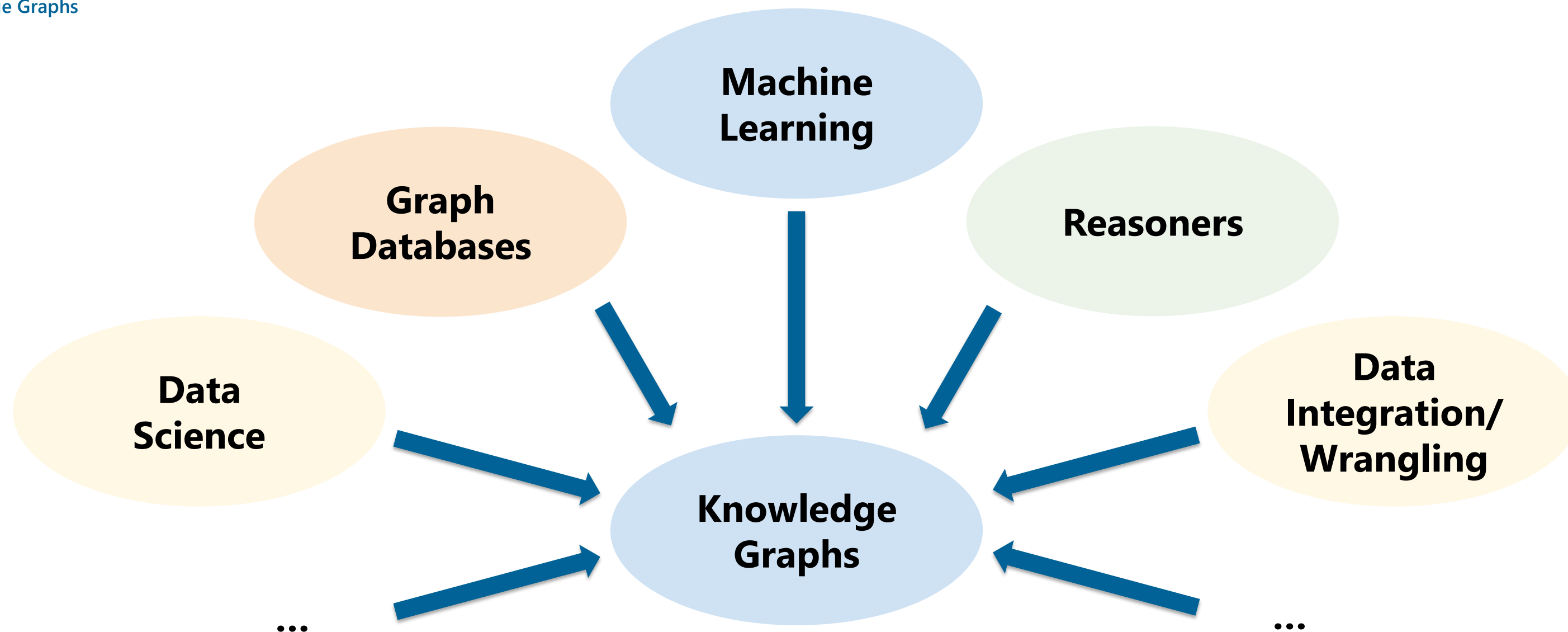


The **Real World**



Theory to **Practice**





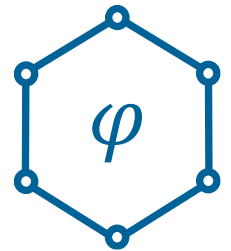


Semantic Web

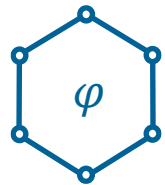
(Relational) Databases

Reasoners

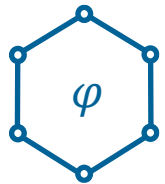
Graph Databases



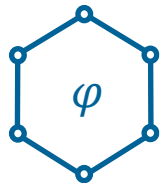
Logical Knowledge in KGs



Creation



Recursion and Creation
A Unified Framework



Recursion and Creation
Warded



The **Real World**



Theory to **Practice**